Microsoft Elevate

CAPSTONE PROJECT

Spotify Music Data Analytics Dashboard using Spotify Streaming Data

PRESENTED BY

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OUTLINE:

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- System Development Approach
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PROBLEM STATEMENT:

Currently, music streaming platforms like Spotify generate a massive amount of data every day related to songs, artists, and user listening behavior. This data includes information such as total streams, playlists, energy level, danceability, and track popularity. However, analyzing such large volumes of data manually is difficult and time-consuming. It becomes challenging to identify trends such as which songs are most popular, which artists are performing well, and how different musical features affect listener engagement. Therefore, there is a need for an analytical dashboard that can visualize Spotify data and provide meaningful insights for better understanding of music trends and streaming patterns.

PROPOSED SOLUTION:

- The proposed system aims to analyze music streaming data and provide meaningful insights through a **Spotify Analytics Dashboard** using data analytics and visualization techniques.
- **Data Collection:**
  Collect historical music data such as track name, artist name, streams, playlists, danceability, energy, and release year from Spotify datasets.
- **Data Preprocessing:**
  Clean the dataset by removing missing values and duplicates, and organize the data for analysis.
- **Data Analysis:**
  Analyze patterns in music streams, artist popularity, and song features to identify trends.
- **Visualization Dashboard:**
  Create an interactive dashboard using **Power BI** to display charts, graphs, and filters for better understanding of music data.
- **Deployment:**
  Publish the dashboard so users can easily access and explore music analytics.
- **Evaluation:**
  Evaluate the dashboard based on clarity of insights and effectiveness of visualizations.
- **Result:**
  The system provides an interactive dashboard that helps users understand music trends, popular artists, and streaming patterns.

SYSTEM APPROACH:

- The **System Approach** describes the method used to develop the **Spotify Analytics Dashboard** for analyzing music streaming data from Spotify.
- **System Requirements**
  - Computer/Laptop with minimum 8 GB RAM
  - **Power BI Desktop** for data visualization
  - Dataset containing Spotify music information (tracks, artists, streams, etc.)
- **Libraries / Tools Required**
  - **Power BI** for dashboard creation
  - **Microsoft Excel / CSV dataset** for data storage and preprocessing
  - Basic data analytics and visualization features available in Power BI to analyze streaming trends.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection:**
  The system uses **data analytics and visualization techniques** to analyze music streaming data from Spotify. The goal is to identify patterns in song popularity, artist performance, and music features.
- **Data Input:**
  Input data includes track name, artist name, total streams, release year, danceability, energy, valence, and playlist appearances.
- **Processing:**
  The dataset is cleaned and analyzed to identify trends and relationships between different music features and streaming counts.
- **Deployment:**
  The dashboard is developed using **Power BI** and published so users can interact with charts, filters, and visual insights about music trends.

RESULT:

- The developed **Spotify Analytics Dashboard** successfully analyzes music streaming data from Spotify and presents meaningful insights through visualizations.
- The dashboard displays key metrics such as **total streams, number of tracks, average danceability, and energy levels**.
- Visual charts help identify **top artists, popular songs, and streaming trends by release year**.
- Scatter plots and bar charts show the relationship between **music features (danceability, valence, energy)** and song popularity.
- The results demonstrate that the dashboard effectively transforms complex music data into **clear and interactive insights for analysis**.

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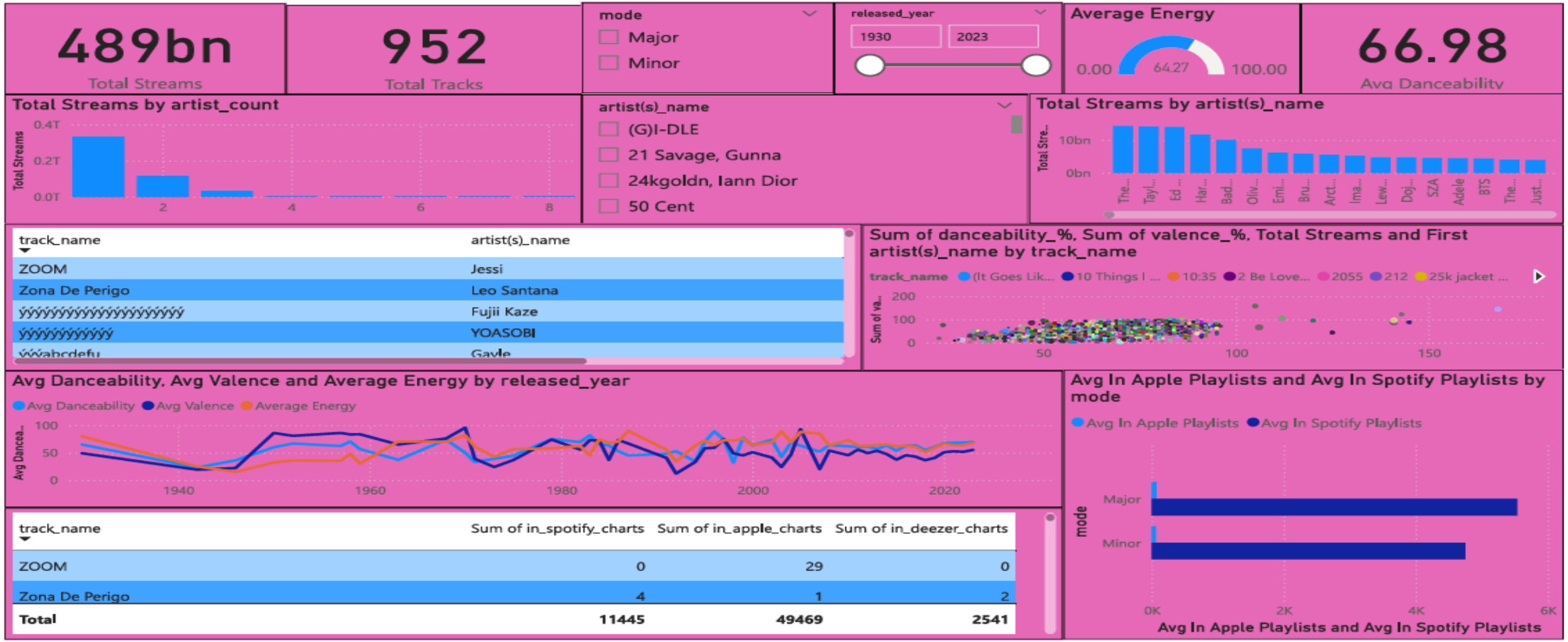
RESULT:


489bn
Total Streams
952
Total Tracks
mode
Major
Minor
released_year
1930
2023
Average Energy
0.00
64.27
100.00
66.98
Avg Danceability
Total Streams by artist_count
Total Streams
0.4T
0.2T
0.0T
2
4
6
8
artist(s)_name
(G)I-DLE
21 Savage, Gunna
24kgoldn, Iann Dior
50 Cent
Total Streams by artist(s)_name
Total Stre...
10bn
0bn
The...
Tayl...
Ed ...
Har...
Bad...
Oliv...
Emi...
Bru...
Arct...
Ima...
Lew...
Doj...
SZA
Adele
BTS
The...
Just...
track_name
artist(s)_name
ZOOM
Jessi
Zona De Perigo
Leo Santana
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CONCLUSION:

The **Spotify Analytics Dashboard** successfully analyzes music streaming data from Spotify and presents it through interactive visualizations. The system helps identify popular songs, top artists, and important music trends. By converting complex datasets into clear charts and insights, the dashboard makes music data easier to understand and analyze. This project demonstrates how data analytics and visualization tools can be used to gain valuable insights from large music datasets.

FUTURE SCOPE:

The system can be improved by adding more data sources from music platforms like Spotify to analyze a larger dataset. Advanced data analytics and machine learning techniques can be integrated to predict song popularity and user listening trends. The dashboard can also be enhanced with real-time streaming data and improved visualizations for deeper insights. Additionally, the system can be expanded into a web or mobile application to make music analytics more accessible to users.

REFERENCES:

- Data Source: Spotify Dataset from Kaggle (Spotify Tracks Dataset).
- Spotify Official Research – Music Streaming Sessions Dataset.
- Research Paper: *Analysis of Machine Learning-Based Music Recommendation System Using Spotify Datasets*.
- Research Paper: *Visual Analytics of Spotify Music Data for Listener Behavior Insights*.

- **GitHub Link:**

Thank You